

RK 1000

SCHOLARLY CONVENTION

Department of Psychology
Dalhousie University

9:30a.m. to 5:00p.m.
Tuesday April 8, and
Wednesday April 9, 1975

Rooms 4258/4263
Life Sciences Centre

TUESDAY APRIL 8, 1975

9:30 Coffee and donuts available

SESSION I CHAIRMAN: JOHN FENTRESS

10:00 C. McNaughton: Circadian Rhythm in Perforant Path Synaptic Efficacy

Granule cell response to perforant path stimulation was found to fluctuate with an apparent circadian rhythm. Since both spike and EPSP components of the evoked potential covaried in time, the effect is probably a synaptic phenomenon. Circadian synapses in the CNS could account for diurnal shifts in behavioral probability patterns.

10:15 Tom Barrett: Agonistic Behavior in the Mongolian Gerbil

Frequency, latency, and duration of open-field agonistic encounters among groups of previously unfamiliar animals decreased as group size increased. Current investigation examines the relevant interanimal cues by placing familiar and unfamiliar groups of different size on opposite sides of opaque, transparent, and open mesh partitions.

10:30 M. Cynader: Abolition of Direction Selectivity in Cat Visual Cortex

Cats were reared in an environment illuminated intermittently by strobe flashes (8 Hz). Striate cortex units in such cats exhibited normal orientation selectivity, but direction selectivity is rarely observed. Preliminary data indicate (1) that direction selectivity is reduced in cat area 18 as well (2) that some recovery occurs following exposure in a normal environment.

10:45 M.G. Yoon: Kinetics of the Topographic Regulation in the Neural Re-connections Between the Whole Retina and a Half-Tectum in Goldfish

If the caudal half of the tectum is excised in adult goldfish, visual projections from the whole retina later becomes orderly compressed onto the remaining rostral half-tectum. It takes at least 33 days for the half-tectum to complete the topographic regulation into a whole.

11:00 General Discussion
Coffee

SESSION II CHAIRMAN: PETER URCUIOLI

30 Daryl Boness: Preliminary Observations on the Grey Seal (Halichoerus grypus) Social System

LATE ADDITION

TUESDAY, APRIL 8,

**12:30 Vincent M. LoLordo, W. J. Jacobs, and Donald D. Foree:
 A failure to block control by a biologically relevant stimulus.**

12:45 Discussion

Lunch

11:45

R.S. Rodger: The Present State of Rodger's Stats

For θ_j a population parameter (mean, proportion, mean-rank, mean vector, regression coefficient), our interest is not in

$$H_0 : \theta_1 = \theta_2 = \dots = \theta_J \quad (1)$$

but contrasts

$$\kappa_h = c_{h1} \theta_1 + c_{h2} \theta_2 + \dots + c_{hJ} \theta_J = \delta_h \quad (2)$$

Traditional critical values $F_{\alpha}; \nu_1, \nu_2$ are inappropriate for (2) and their power drops rapidly with ν_1 (non-significant interactions are often an artifact). Tables of $F[F_{\alpha}]; \nu_1, \nu_2$ for (2) are in press.

12:00

Adam Carr: Effects of Shock and Shock-Associated Stimuli on Alert-Posturing in the Mongolian Gerbil

Walters and Glazer (1971) have shown that alert-posturing in the gerbil may be facilitated by response-contingent presentations of a shock-predictive stimulus. The present experiments demonstrate that alert-posturing can also be facilitated by a stimulus which predicts a shock-free period, or suppressed by response-contingent shock. Facilitation of alert-posturing by a shock-predictive stimulus depends upon an interaction between the amount of training which precedes testing and the locus of testing.

12:15

W.W. Martin: The Effect of Acquired Affective Value on Visual Fixation

Twenty-four subjects were given an aversive conditioning procedure in which human face served as conditioned stimuli. Duration and frequency of visual fixations were subsequently recorded for pictures of the conditioned stimuli. Results failed to show an effect for affective value. Discussion speculates on stimulus influences as explanations for spontaneous attention.

12:30

General Discussion
Lunch

SESSION III

CHAIRMAN: FRANK GOODKIN

1:30

J.C. Fentress: Functional Integration of Stereotyped Movements

By examining individual components and their sequential coupling in relatively stereotyped movement sequences it is possible to construct precise models of dynamic integration. In particular control systems become more tightly and centrally organized as activation levels or output demands increase. A boundary-state model of interactive self-organizing control systems is proposed.

1:45 F. Wilkinson: Reversal of Monocular Deprivation Within The
Critical Period - Behavioral Effects

Kittens were given 11 days of monocular visual experience beginning on day 26 of life. This was followed by either a) reverse suturing and 11 days of experience with the initially deprived eye later in the critical period, or b) no further experience until 4½ months of age. Interocular differences in acuity were assessed behaviorally in adulthood.

2:00 F. Baker: Morphological Changes in the Laminar Structures of the
Goldfish Optic Tectum Following Surgical Operations

There are six distinct horizontal layers in the optic tectum of normal goldfish. After the removal of ½ tectum there are morphological changes in the stratum opticum (2nd) and stratum fibrosum et griseum superficiale (3rd) layers which becomes one with the appearance of nerve groupings or bundling. Because these layers are the main target zone of retinotectal projection it would suggest that the input from the eye accounts for this reorganization.

2:15 Bruce McNaughton: Transport and Turnover of Proteins Undergoing
Axonal Transport in the Teleost Optic Nerve:
Effects of Complex Visual Stimulation

After intraocular injections of labelled precursors, complex visual stimulation altered the distribution of labelled protein in the Trout optic system. The results are interpreted in terms of a model based on current understanding of the role of axonal transport in synaptic transmission.

2:30 R. Adamec: Effects of Food Deprivation on Mouse Killing Behavior
of Rats

Hunger potentiates several oral-olfactory and tactual behaviors toward prey. With repeated kills, latency to kill and frequency of all behaviors except biting decline. This decline does not appear in rats which do not kill in response to deprivation. Moreover, defensive responses of both prey and predator interfere with the facilitation of attack.

2:45 General Discussion
Coffee

SESSION IV CHAIRMAN: R.S. RODGER

3:15 G.V. Goddard: Must the Post-Synaptic Cell Respond During
"Learning" before "Memory" is Created in a Synapse
Of the Monkey Hippocampus?

I don't know, but I am trying to find out. I think it is important because it is possible to imagine reasonable neuro-physiological explanations of learning and memory if lasting synaptic enhancements are response contingent. It is very difficult if lasting synaptic potentiation results from mere pre-synaptic activity.

3:30 J.A. Stevenson: Tectal Neogenesis and its Response to Tectum Denervation

Orderly reinnervation of goldfish optic tectum depends, in part, upon properties inherent in the tectum. An experiment is designed to study the temporal sequence of new cell formation in tectum following denervation, and its relation to normal neogenesis.

3:45 D. Mitchell: A Case of Meridional Aniseikonia

4:00 R. Dale: Interaction of "Schedule-Relevant" and "Schedule-Irrelevant" Activities on a Fixed-Interval Reinforcement Schedule

Six pigeons obtained food by key pecking on a fixed-interval reinforcement schedule. Water was freely available in the test chamber. The pigeons paused (did not key peck) after obtaining food, and occasionally drank during this pause. Drinking increased the pause to key pecking, but did not effect the subsequent key peck rate.

4:15 C. Adamec: Sex Differences in Response to Marijuana in a Social Setting: Or, You Get High With a Little Help From Your Friends But You Have More Fun if You are a Woman

4:30 J.A. McNulty: The Growth of a Percept with Changing Stimulus Features

4:45 General Discussion
Social Hour

WEDNESDAY APRIL 9, 1975

9:30 Coffee and donuts available

SESSION V CHAIRMAN: ADAM CARR

10:00 S. Nakajima: Hormonal Factors Involved in Passive Avoidance

To investigate how corticosteroids influence passive avoidance, I assayed the concentration of the steroids at various stages of avoidance testing, and then examined avoidance behavior under pharmacological manipulation of various hormones. Corticosteroids appear to influence avoidance behavior by modifying the sympathetic activity.

10:15 P. Urcuioli: Oddity-From-Sample Performance in Pigeons

Pigeons were trained on a modified three-key oddity-from-sample procedure. The birds reported odd stimulus pairs by pecking odd comparison hues, and reported matching stimulus pairs by refraining from pecking matching comparisons for 5 sec. Acquisition data and preliminary transfer data portray highly variable performances.

10:30 R. Sutherland: Changes in Lateral Hypothalamic Self-Stimulation Following Ventromedial Hypothalamic Damage

Bilateral electrolytic lesion in VMH of rats trained to respond for electrical stimulation in LH, causes a large increase in responding shortly after lesioning. On subsequent days this increase disappears, responding declines below baseline and recovers two weeks post-lesion. Loss of VMH inhibition of LH may underly these changes.

10:45 P. Anderson: Interocular Transfer of Directionally Selective Threshold Elevation

11:00 General Discussion
Coffee

SESSION VI CHAIRMAN: TOM BARRETT

11:30 Ilan Golani: A Study of the Ontogeny of Grooming Behavior in Infant Mice Through High Speed Filming and Movement Notation Analysis

11:45 Georgia Cottrell: Hippocampal Corticosteroid Injections Counteract Cycloheximide-Induced Amnesia

The amnesic effect of cycloheximide has been shown by Nakajima to be linked to adrenal corticosteroid function, not inhibition of brain protein synthesis. Subcutaneous or intrahippocampal injections of a corticosteroid counteracted the cycloheximide effect on retention in a passive avoidance task, suggesting that corticosteroids act on hippocampal neurons to alter subsequent behavior.

12:00 Frank Goodkin: Some Thoughts on Learning

Experiments designed to study animal learning and memory frequently consist of operant and/or classical conditioning. Both procedures are relatively complex, special cases of learning. Some of the generalizations drawn from these procedures may be misleading to our understanding of learning. Some thoughts about a procedure designed to measure a simpler, more general form of learning will be presented.

12:15 W.J. Jacobs: The Relation of a Compound Stimulus to Shock Determines the Direction of Attention

When a compound auditory-visual stimulus precedes barpress avoidance responding, the auditory element controls a subsequent decrease in wheelturn avoidance behavior. When the compound follows the barpress avoidance response, the visual element controls a subsequent decrease in wheelturn avoidance responding.

12:30 General Discussion
Lunch

SESSION VII CHAIRMAN: SHINSHU NAKAJIMA

1:30 Vern Honig: On the Functional Equivalence of Stimuli that Signal Different Aversive Conditions

Pigeons are taught two independent, orthogonal discriminations, one between random-interval DRL reinforcement schedules, and the other between reinforcement and extinction. They learn to reduce time spent in DRL by making a changeover response. This strategy transfers immediately to the extinction condition as soon as the changeover contingency is introduced. The pigeon appears to treat DRL and extinction conditions as functionally similar or equivalent.

1:45 R. Brown, R.S. Rodger, R. Rosebrugh: The Analysis of Behavioral Sequences by Means of Formal Grammars

A right-linear, probabilistic grammar is used to describe the structure of the rhythmic song of the wood pewee. One advantage of this method over contingency χ^2 and information-theory is that absent sequences constitute no problem. The grammar method produces rules (each with a theoretical probability) to generate behavioural sequences, with checkable probabilities, for comparison with observations.

2:00 Ray Blanchard: The Surprise Value of Food Determines its Effectiveness as a Reinforcer

One group of pigeons (Positive) received keylight - food pairings in a colored houselight previously associated with food, one (Negative) in a houselight previously associated with extinction, and one (Novel) in an unfamiliar houselight. The Negative group responded most to the keylight - CS and the Positive group responded least.

2:15

H. James, R.W. Dykes: A Harbour Seal's Compass

Normal seals released at unfamiliar places twenty-four hours after capture steer a fixed compass course to the sea. Anosmia, translucent blindfolds, middle ear lesions do not affect this behavior. Optic nerve section abolishes it; inner ear lesions make seals unable to distinguish North from South; blindfolded deaf seals have no map.

2:30

R. Klein: The Irony of Visual Dominance

Information presented visually frequently controls the subjective reports and performance of human subjects. This visual dominance is not "wired-in". It results from a bias (not necessarily conscious) to selectively attend vision. We (humans) may adopt this bias to overcome the relatively inferior alerting capability of visual stimuli.

2:45

General Discussion
Coffee

SESSION VIII

CHAIRMAN: ROBERT DALE

3:15

Jim Day: Effects of Acoustic and Visual Confusability on Right and Left Hemisphere Performance of a Letter Matching Task

Previous work has suggested that the left and right hemispheres of the human brain are specialized for performing verbal and non-verbal tasks respectively. An experiment was designed to determine whether hemispheric performance differences are attributable to different degrees of specialization in the two hemispheres or to absolute lateralization of function.

3:30

G. Moran: Stranger in a Strange Land

The literature to date on the social behavior of the wolf consists largely of well-informed conjecture relating to functional and causal questions. The few attempts at description have employed the static "snapshot" approach and are devoid of contextual information. An adequate understanding of the mechanisms of social interactions first requires a descriptive system, such as the Eshkol-Wachmann Movement Notation, which deals with the dynamic nature of the behavioral system.

3:45

R.M. Douglas: Electrophysiological Evidence for the Modifiability of Synaptic Transmission

In the rat hippocampus a long lasting change in synaptic transmission can be produced by stimuli that mimic the natural discharge pattern of single units. This potentiation occurs whether or not the output of the post synaptic cells is blocked by inhibition produced by a prior input.

4:00

A.L. Riley: Chlordiazepoxide-Induced Reversal of an Amphetamine-Established Aversion: Non-Associative Effects

Cappell and LeBlanc (1973) reported the reversal of an amphetamine-established aversion by chlordiazepoxide (CDZP). These results were explained in terms of the anti-punishment effect of CDZP on amphetamine-established behavior. The present report, however, suggests that this reversal is a function of the interaction between increased thirst induced by CDZP and the insensitivity of a one-bottle test in measuring conditioned taste aversions.

4:15

Paul A. Nau: A Study of the Directed Nature of Avoidance Responding

Rats were trained to shuttle either toward or away from a pre-shock stimulus. In general, rats required to run away learned the avoidance response while those required to run toward did not. However, in a smaller, unilluminated chamber the opposite results were obtained. The results suggest that environmental stimuli unrelated to an avoidance contingency influence initial response probability.

4:30

G. Chernenko: Another Look at Development of Ocular Dominance Distribution in Cat Superior Colliculus

4:45

General Discussion
Party